Automated Risk-Scoring System (Framework) for Financial Institutions

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Problem Statement

Payment flows

- WIRE • SEPA • SWIFT
- Cash deposit/withdrawal
- Currency exchange
- Daily high volume

Detection models

- Risk scoring / classification
- Rules + ML
- Thresholds & calibration

Alerts and investigation

- Alert queue for analysts
- Case review & escalation
- Regulatory reporting

Operational reality

- Low base rate (≈1% suspicious)
- High FP → heavy workload
- Need explainable decisions

Evolving typologies

- **New fraud/AML patterns appear**
- Scenario overlap increases
- Concept drift over time

Data constraints

- **Limited labelled cases**
- Privacy restrictions
- Reproducible benchmarking is difficult

- Problem: Evaluate detection that stays reliable under high volume, class imbalance, and changing scenarios, without using real customer data.
- Research Object: Procedures for risk assessment and detection of suspicious transactions in financial institutions.
- Research Subject: Scenario-based synthetic transaction data generation and classification for suspicious activity detection and scenario typing.

- Research Goal:
  - Develop and experimentally validate a demonstrative prototype of an automated risk-scoring framework.
- Hypothesis:
  - Synthetic transaction data created through scenario and rule-based methods is suitable for training ML models that automate risk assessment in the financial sector
- Research questions:
  - Can realistic synthetic data replicate both normal and suspicious behaviours, including complex fraud schemes?
  - What technical and regulatory challenges arise when creating such data, and how can quality be evaluated?
  - How well do ML models trained on multiple synthetic data packages adapt to evolving fraud strategies?

Literature Review and Research Gaps

Records identified **(Scopus, ScienceDirect, Consensus)** (*n* = ***3122***) → **Excluded:** irrelevant, duplicate, or lacking sufficient quality (*n* = ***3040***) → Shortlisted for synthesis (*n* = 82) → Excluded after evidence extraction (*n* = 22) → **Included in the final research** (*n* = ***62***)

Key limitations

- **High volume**
- **High false positives**
- **Explainability** required
- Limited real-world data
- Limited control over explicit AML scenarios/typologies
- Multi-dataset / drift evaluation not standardised

Research gaps

- No standard **reproducible benchmark**
- **Synthetic realism** is rarely calibrated
- **Real-data** studies are hard to reproduce
- Explainability is limited
- Needs enough real data
- Weak scenario control
- Multi-datasets **evaluation** is not standard
- Hard to guarantee **scenario coverage**

(Oztas et al., 2024; Kute et al., 2021; Altman et al., 2023; Alexandre & Balsa, 2023; Chen et al., 2021; Altman et al., 2023; Kute et al., 2021)

Research Limitations

- Real data: **<500** rows
- Synthetic data: **10,000/100** and **50,000/500** (normal/suspicious **≈1%)**
- **Privacy/compliance** restricts real-data reuse
- **Scenario evolution:** the scenario catalogue expands over time
- Recall **≥0.98** (primary)

Framework Design

1. Choosing a data synthesis method

- Model and/or rule-based approach
  - Limitation: up to <500 real rows
  - Requirements: adaptable and updatable scenarios

2. Scenario catalogue

- Sources: FI guidelines and EU directives
- Grouping by mechanism; scenario parameters

3. Data synthesis

- Model background + scenario "overlays"
- Rule-based simulation by specifications

4. Data quality check

- Rule-based validation
- Statistical feature profiling

5. Class balance

- SMOTE / ADASYN / ROSE
  - Class weights
  - Focal loss

6. Model selection

- AutoML with feature analysis for ranking.
- Expert validation and shortlisting.

7. Training mode

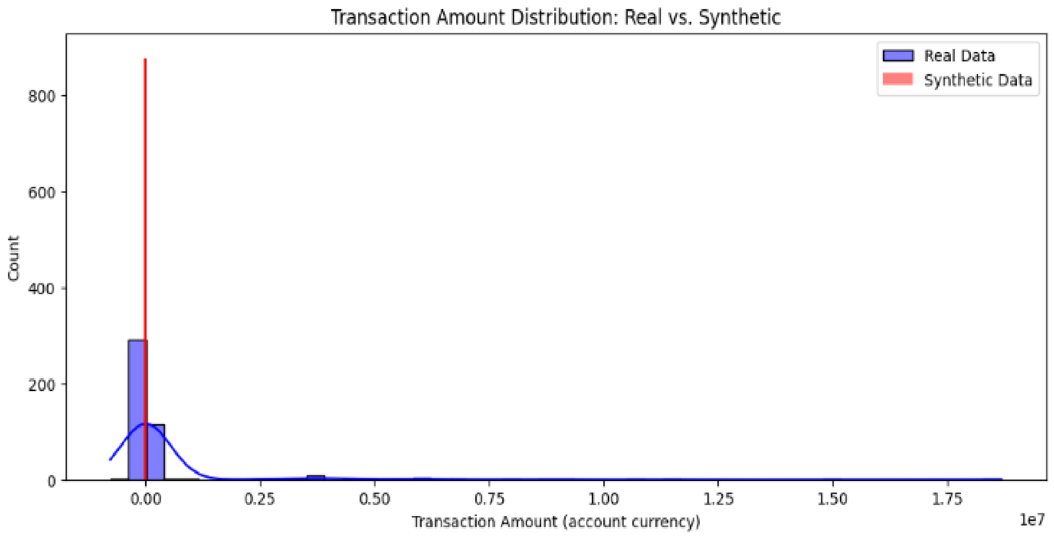
- Incremental/full retraining
- Update rules, thresholds, and monitoring

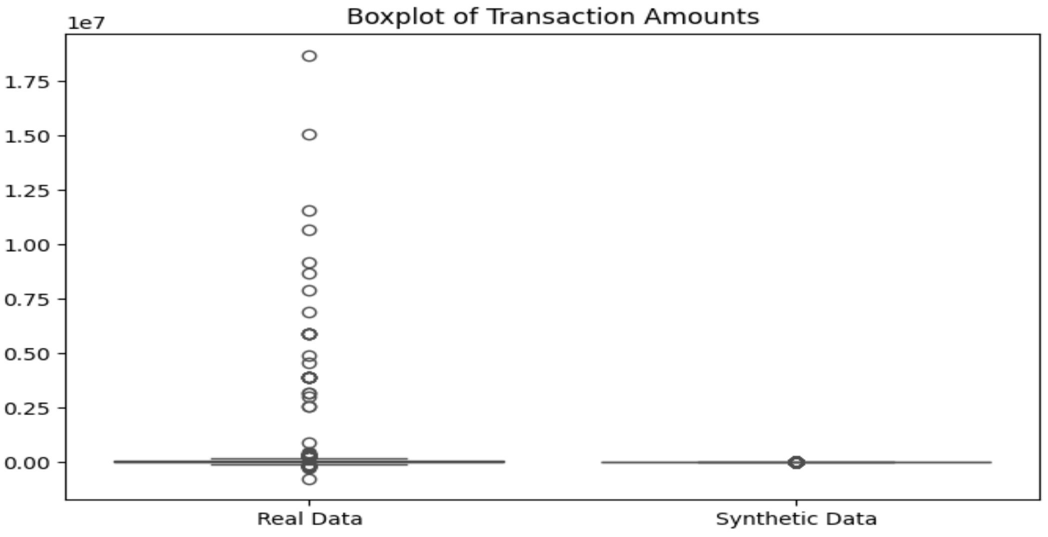
8. Adaptivity check

- Data Packages
- Evolving Scenarios

9. Conclusion

GAN-Based Synthetic Data





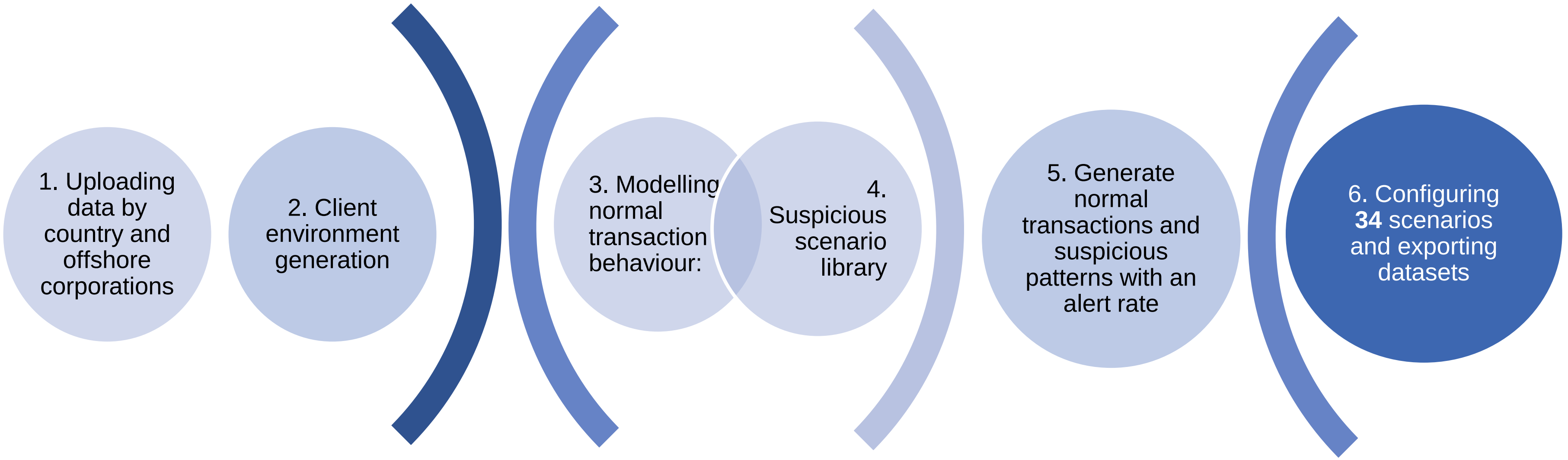
Extreme values are not preserved in synthetic data

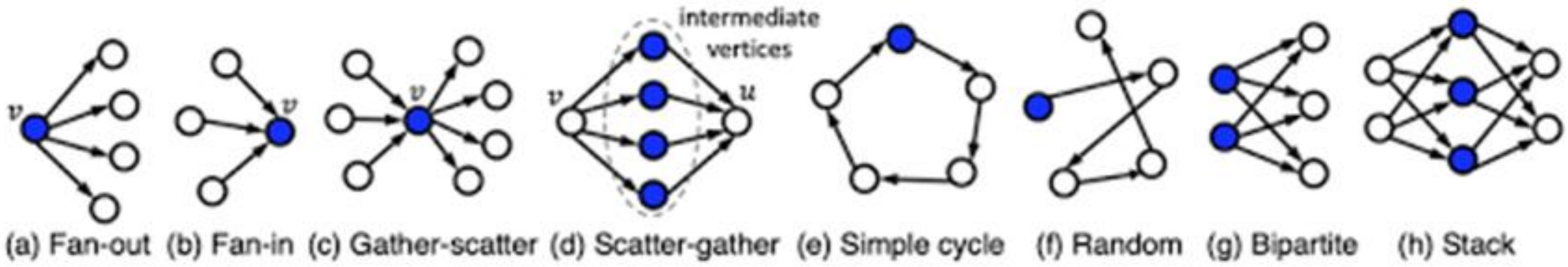
- **Results**
  - Amounts: **strong mismatch** in the upper tail ($KS = 0.66$, $p \ll 0.001$)
  - Categories: no significant difference (χ^2 $p \approx 1.0$)

GAN preserves categorical proportions but *fails* to reproduce heavy-tailed transaction amounts

- Not suitable for fraud detection benchmarking

Rule-Based Synthetic Data





1. Transaction flow and velocity monitoring
- Transit Payments
- Daily/Monthly Turnover Thresholds
- Circular Money Flow
- Rapid Currency Conversion
- Transaction Structuring
- Sudden Activity After Dormancy
- Newly Incorporated Entity High Volume
- Unusual Transaction Speed
- Identical Transaction Size

2. High-risk jurisdiction detection
- High-Risk Country Daily/Monthly Transactions
- Very High-Risk Country Transactions
- First-Time High-Risk Country Activity
- Prohibited Country Transactions
- Foreign Country Transaction Ratio
- Restricted Country Currency

3. Transaction pattern analysis
- Balanced Flow Analysis
- One-to-Many Pattern
- Many-to-One Pattern
- Round Amount Transactions
- Historical Transaction Spike
- Unusual Increase in Turnover
- Foreign Currency First

4. Entity and relationship risk
- Shell Company Transactions
- Multiple High-Risk Transactions

5. Contractual and documentation issues
- Insufficient Transaction Description
- Minimal Payment Details
- Repetitive/Missing Reference Numbers
- Inconsistent Business Activities
- Fictitious Loan Transactions
- Third-Party Financing

6. Cash and currency operations
- Large Cash Withdrawals
- Abnormal Cash Activity
- Cash Withdrawal Pattern Changes
- Multiple Currency Transactions

Syn100: 5 packages (**B1–B5**) × (10,000 normal + 100 suspicious); **Syn500**: 5 packages (**B1–B5**) × (50,000 normal + 500 suspicious).

Model Selection

Manual

Model family	Interpretability	Class imbalance	Batch-wise update	Data scale	Limitations
Gradient-Boosted (GBM, XGB)	✓ (post-hoc SHAP)	✓ (class-weights / focal / scale pos weight)	✓ (batch updates; not online)	5 k – 1 M	Needs careful probability calibration in tail regions; sensitive to feature leakage.
Random Forest (RF)	± (feature importance / partial interpretability)	± (class weights / sampling)	✓ (batch updates; not online)	≥ 1 k	Lower quality ceiling than GBM.

Legend: ✓is a strength; **±** is an acceptable/tuning dependent; ✗is a weakness; k is a thousand.

Note: “Batch-wise update” refers to model maintenance using new data batches and retraining, not true online learning.

H2O AutoML

1	**XGB**
2	**GBM**
3	**GBM**
4	**GBM**
5	**XGB**

- **Used as an independent benchmark**
- **Across 5 data packages, GBM/XGB consistently ranked as leaders**
- **Confirms manual model selection**

The gradient-boosted model family (GBM/XGB) was selected as the primary performance-oriented approach, while Random Forest was used as a robust baseline for comparison.

Balancing methods

Proportion of suspicious transactions

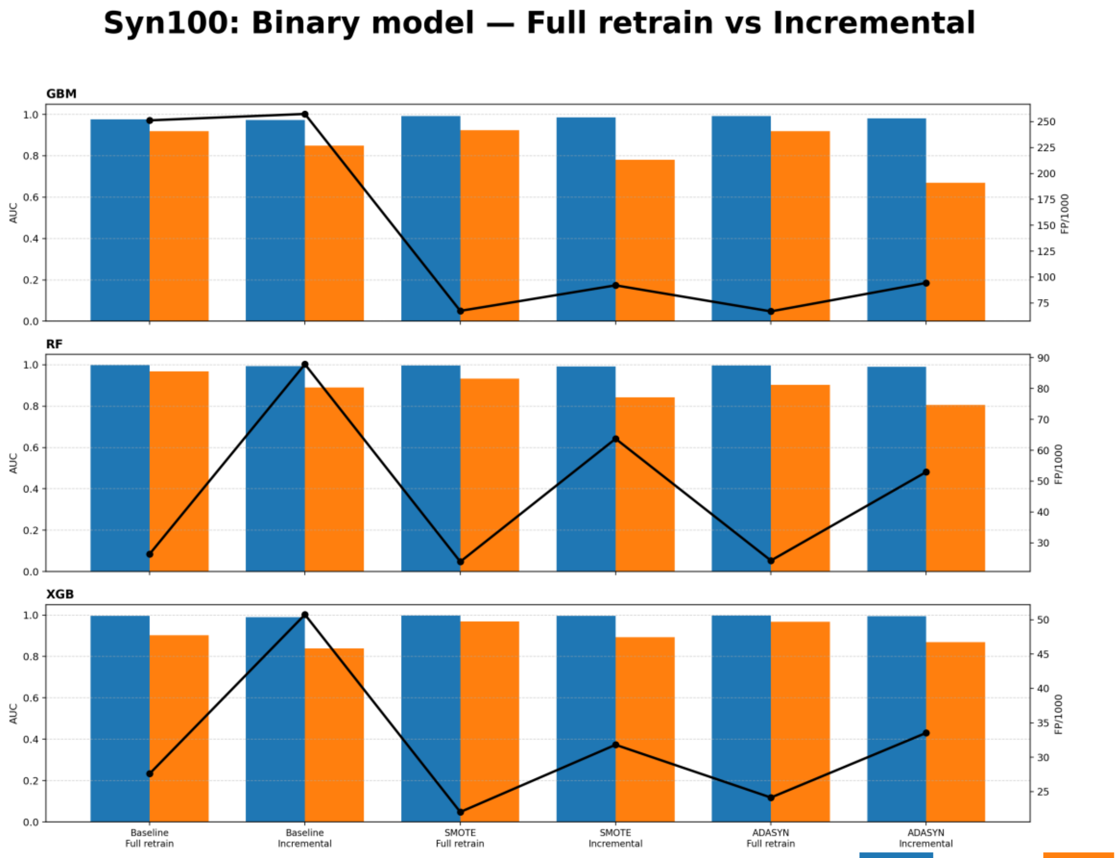
		Baseline	SMOTE	ADASYN	ROSE
	Suspicious transactions	65	4007	4007	8015
Syn100	Normal transactions	8015	8015	8015	8015
	Proportion	0.008	0.3333	0.3333	0.5
	Suspicious transactions	1652	19374	18815	38748
Syn500	Normal transactions	38748	38748	38748	38748
	Proportion	0.0409	0.3333	0.3269	0.5

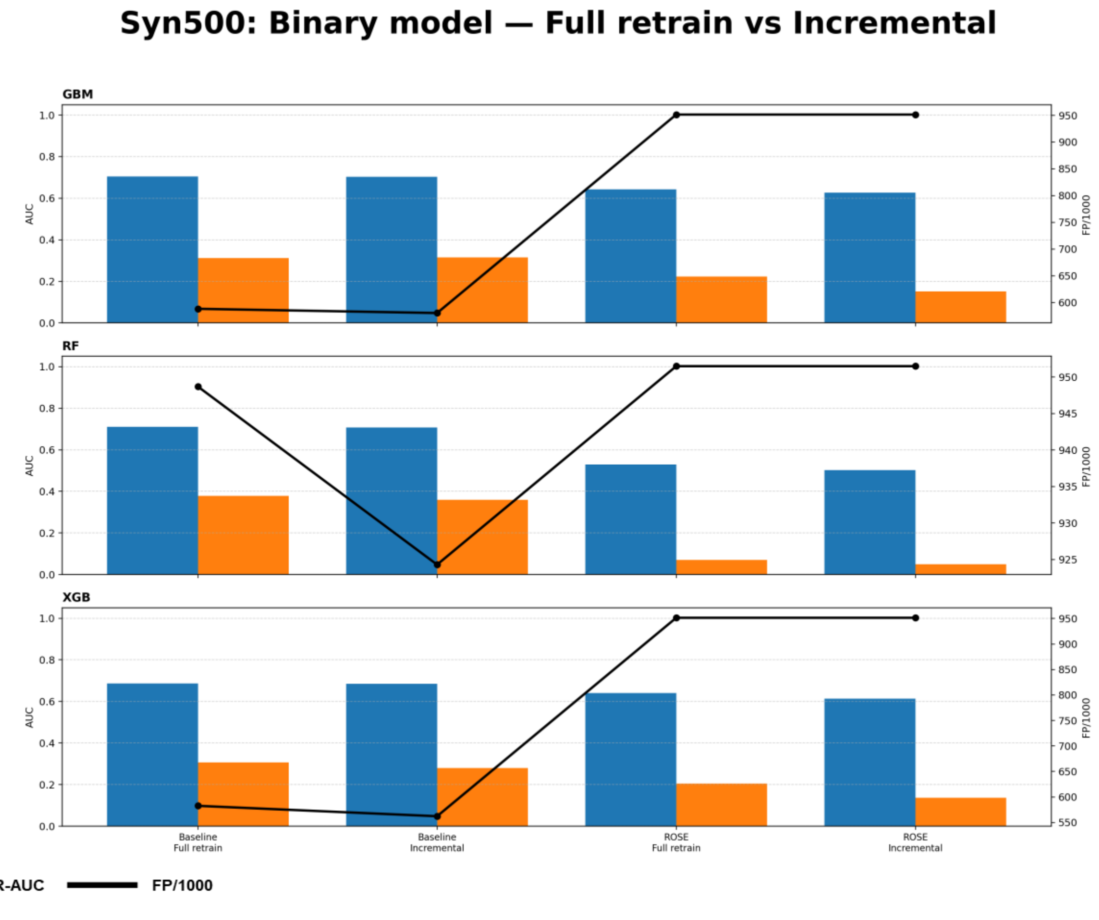
Quality assessment		**Syn100**		**Syn500**	
Variant	**Algo**	**Precision**	**FP/1000**	**Precision**	**FP/1000**
Baseline	GBM	0.978	0.396	0.070	629.366
Baseline	RF	0.975	0.495	0.052	857.485
Baseline	XGB	0.986	0.198	0.068	656.356
SMOTE	GBM	0.994	0.099	0.053	832.574
SMOTE	RF	0.975	0.495	0.051	879.386
SMOTE	XGB	0.994	0.099	0.057	767.604
ADASYN	GBM	0.981	0.297	0.052	849.822
ADASYN	RF	0.967	0.693	0.051	882.475
ADASYN	XGB	0.969	0.495	0.057	777.545
ROSE	GBM	0.989	0.198	0.066	668.693
ROSE	RF	0.989	0.198	0.052	864.990
ROSE	XGB	0.994	0.099	0.071	620.337

- Syn100 dataset:
  - Baseline + SMOTE + ADASYN (exclude ROSE)
- Syn500 shortlist:
  - Baseline + ROSE (exclude SMOTE/ADASYN)
- Baseline:
  - no-balancing reference (zero indicator)

Purpose: shortlist configurations for binary detection (**Recall ≥ 0.98** enforced at calibration).

Binary model (Full retrain vs Incremental)

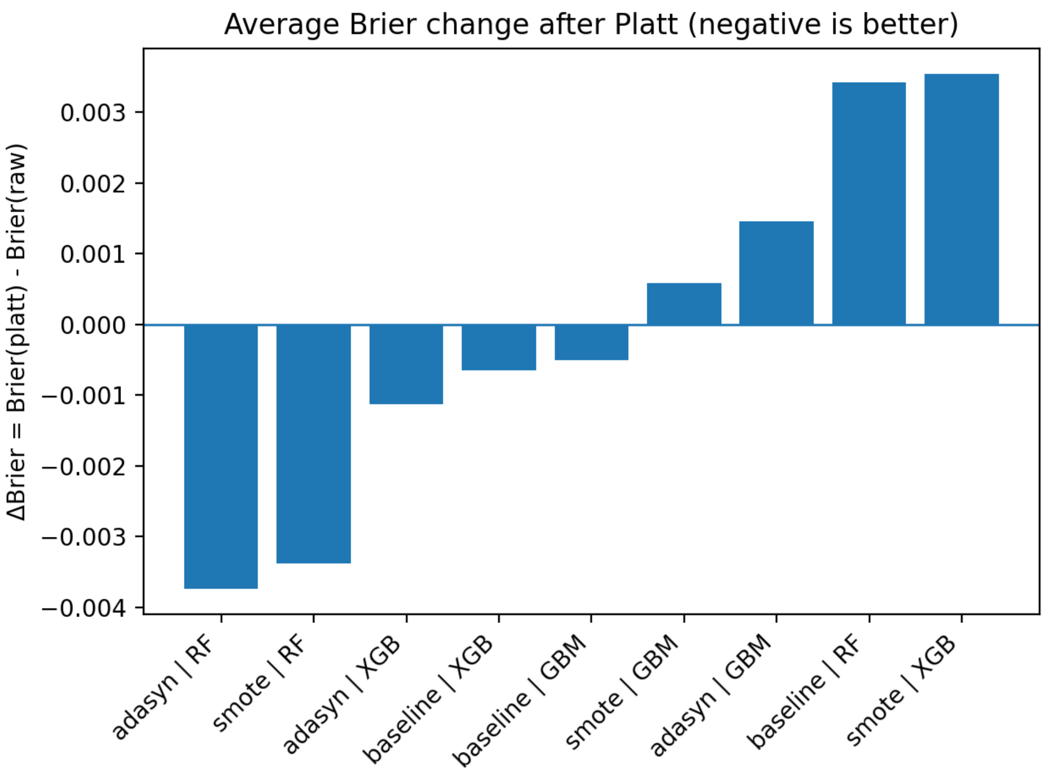




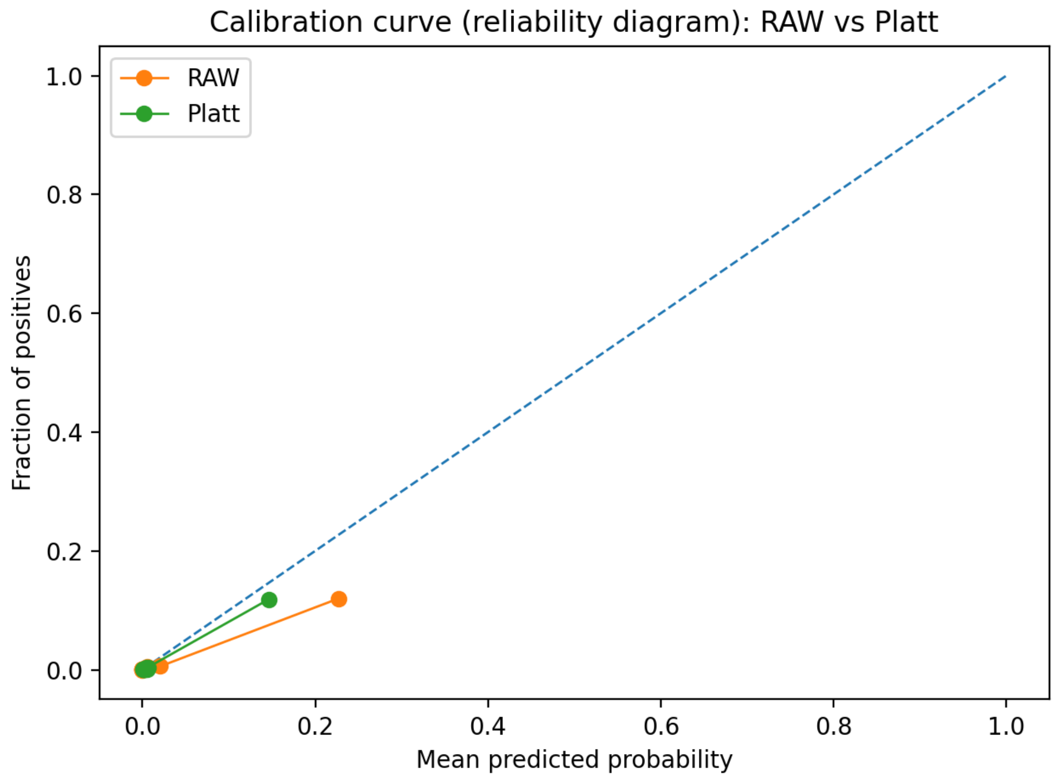
Model	Mode	Best variant	ROC-AUC	PR-AUC	FP/1000
GBM	**Full retrain**	**ADASYN**	0.992	0.920	66.8
RF	**Full retrain**	**SMOTE**	0.997	0.932	23.9
XGB	**Full retrain**	**SMOTE**	**0.998**	**0.970**	**22.0**
XGB	**Full retrain**	**Baseline**	0.996	0.903	27.6

Model	Mode	Best variant	ROC-AUC	PR-AUC	FP/1000
GBM	**Full retrain**	**Baseline**	0.704	0.312	**588.2**
RF	**Full retrain**	**Baseline**	0.710	0.378	**948.7**
XGB	**Full retrain**	**Baseline**	0.687	0.306	**582.6**

Out of Fold and Platt calibration Syn100



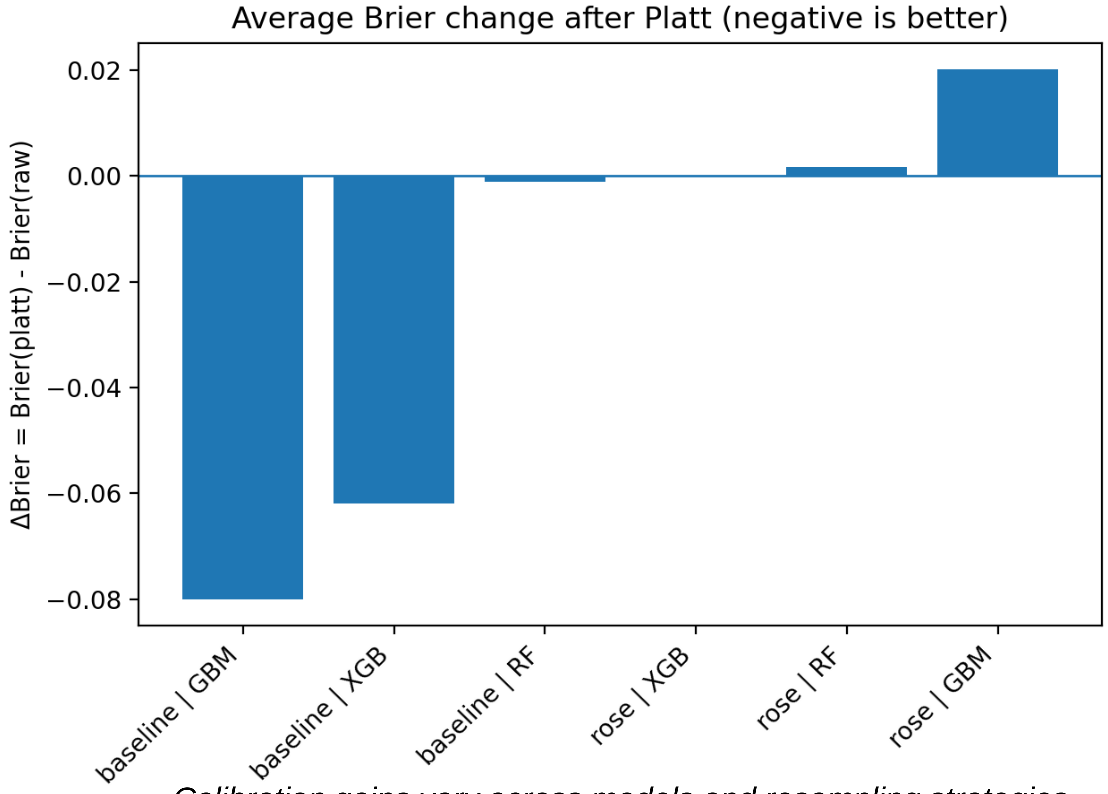
Negative ΔBrier indicates improved calibration.



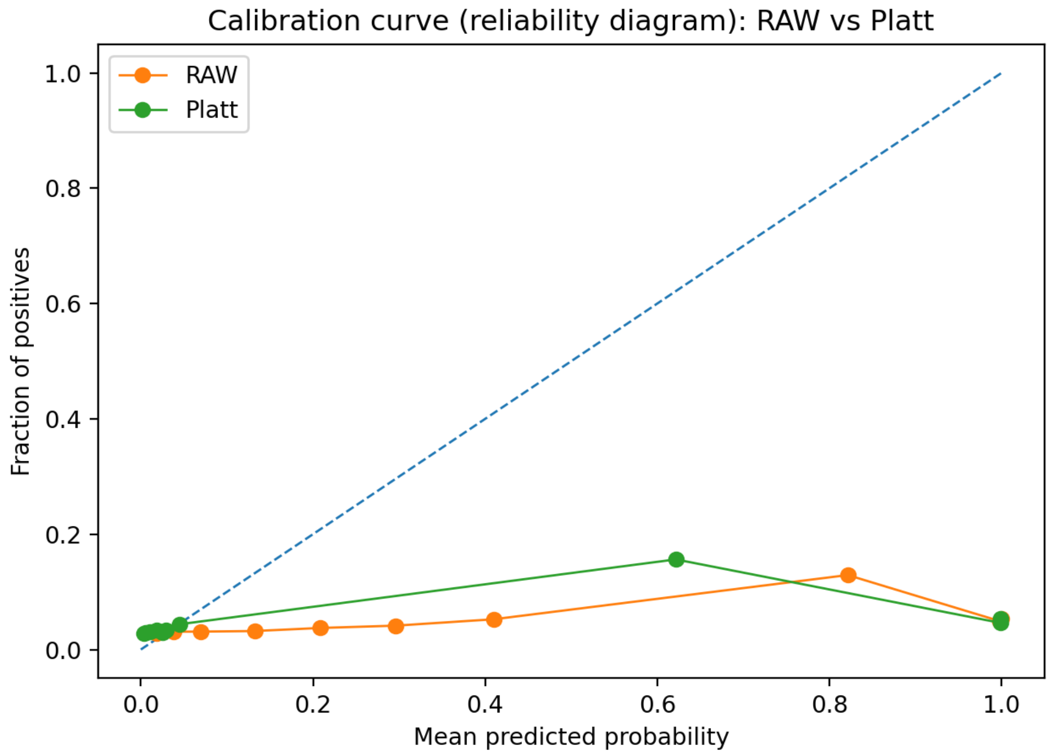
Platt-scaled probabilities move closer to the ideal calibration line.

- OOF predictions provide unbiased probability estimates for calibration
- Platt scaling improves probability calibration (lower Brier score)
- Ranking performance (ROC/PR) is preserved; only probability reliability changes

Out of Fold and Platt calibration Syn500



Calibration gains vary across models and resampling strategies.



Platt scaling changes the shape of the calibration curve across probability ranges.

- Platt scaling improves calibration mainly for baseline GBM/XGB models
- Calibration gains are model and sampling-dependent
- Probability reliability improves, while ranking behaviour remains unchanged

Multiclass model Syn100 Baseline

Data	Model	Accuracy	F1	Precision	Recall	TopK1	TopK2	TopK3
B1	GBM	0.34286	0.40741	0.50000	0.38095	0.34286	0.48571	0.48571
B2	GBM	0.60714	0.58185	0.53497	0.75000	0.60714	0.67857	0.67857
B3	**GBM**	**1.00000**	**1.00000**	**1.00000**	**1.00000**	**1.00000**	**1.00000**	**1.00000**
B4	GBM	0.89655	0.88474	0.91296	0.88889	0.89655	0.89655	0.93103
B5	GBM	0.96667	0.90351	0.87500	0.97500	0.96667	0.96667	0.96667
B1	RF	0.42857	0.34354	0.34286	0.38776	0.42857	0.45714	0.51429
B2	RF	0.89286	0.87719	0.87500	0.93182	0.89286	0.96429	1.00000
B3	**RF**	**0.95000**	**0.93333**	**0.94444**	**0.94444**	**0.95000**	**1.00000**	**1.00000**
B4	RF	0.86207	0.87525	0.92460	0.88889	0.86207	0.93103	0.96552
B5	RF	0.76667	0.52778	0.53125	0.70000	0.76667	1.00000	1.00000
B1	XGB	0.28571	0.26154	0.23636	0.40000	0.28571	0.37143	0.51429
B2	XGB	0.57143	0.49684	0.42013	0.66667	0.57143	0.78571	0.96429
B3	**XGB**	**1.00000**	**1.00000**	**1.00000**	**1.00000**	**1.00000**	**1.00000**	**1.00000**
B4	XGB	0.89655	0.90194	0.90370	0.91667	0.89655	0.93103	0.96552
B5	XGB	0.76667	0.52778	0.53125	0.70000	0.76667	1.00000	1.00000

Multiclass model Syn500 Baseline

Data	Model	Accuracy	F1	Precision	Recall	TopK1	TopK2	TopK3
B1	GBM	0.95806	0.70919	0.70535	**0.71960**	0.95806	0.97572	0.98455
B2	GBM	0.83726	0.43750	0.71259	0.38291	0.83726	0.89293	0.89936
B3	GBM	0.90223	0.51649	0.57894	0.54366	0.90223	0.92624	0.96741
B4	GBM	0.93605	0.40353	0.45845	0.38338	0.93605	0.95155	0.95543
B5	GBM	0.83863	0.41852	0.48986	0.41378	0.83863	0.89976	0.92421
B1	RF	0.81457	0.52321	0.67952	**0.52566**	0.81457	0.99338	1.00000
B2	RF	0.86081	0.55444	0.73247	0.50592	0.86081	1.00000	1.00000
B3	RF	0.80961	0.09501	0.18380	0.08332	0.80961	0.99314	0.99828
B4	RF	0.88178	0.11876	0.16014	0.10648	0.88178	0.94767	0.96512
B5	RF	0.75795	0.06310	0.08787	0.06250	0.75795	0.87042	0.90220
B1	XGB	0.77263	0.35048	0.52040	0.35807	0.77263	0.97792	1.00000
B2	XGB	0.82655	0.36808	0.45538	0.32741	0.82655	0.99358	1.00000
B3	XGB	0.93139	0.65993	0.71978	**0.64094**	0.93139	0.97770	0.98971
B4	XGB	0.92442	0.32736	0.36743	0.32017	0.92442	0.95349	0.97674
B5	XGB	0.87042	0.36579	0.37505	0.41808	0.87042	0.91198	0.93888

Conclusion

- The hypothesis is partly confirmed: rule-based synthetic data can be used to train ML detectors for risk screening, but it is insufficient for fully automated risk decision-making.
- Q1 (behaviour realism): normal and suspicious behaviours are reproduced with realistic overlap; as a result, when a strict recall target is applied, precision decreases due to the lack of a clear separation between classes.
- Q2 technical and regulatory constraints (privacy restrictions and limited labelled real data) require fully synthetic generation, while quality is evaluated via privacy–utility–fidelity: privacy is preserved, utility is high for controlled training/benchmarking, and fidelity is limited by predefined rules and simplified business logic.
- Q3 (adaptation and scenario typing): binary detection remains stable across data packages and keeps high recall under both incremental updates and full retraining, with only a small quality difference between these regimes. This suggests the model can be updated in batches without major degradation. When a new scenario appears, the binary model can still flag transactions as suspicious, but it cannot assign a reliable scenario label, so scenario typing is mainly useful for triage rather than final labelling.

Scope and Contributions

Scope	Contribution
Synthetic data with **34 AML scenarios**, incl. **graph-style patterns**	Focus on **realism impact**, not “maximum accuracy”
Two datasets: **Syn100 (10k/100)** + **Syn500 (50k/500)**, each with **5 data packages**	“Too clean” synthetic data → **near-perfect metrics** can be misleading
Binary detection: GBM / XGBM / RF + **H2O AutoML** benchmark	More realistic AML-like data → **class overlap** → higher **FP/FN** (closer to operations)
Multiclass typing: on top of binary outputs using **Top-K**	Multiclass stacking supports **triage**, but labels are **moderately reliable**
Imbalance handling (binary): **Syn100** → **Baseline/ADASYN/SMOTE; Syn500** → **Baseline/ROSE**	At **Recall = 0.98**, precision can drop to **~4–6%** → needs ranking / filters
Training & evaluation: **Incremental vs Full retrain**; final binary with **OOF + plateau**	Oversampling + retraining tested; **incremental ≈ full retrain** (similar stability)
Total tests = 75 (binary) + 30 (multiclass) = 105	GAN on ~500 real rows: good on “typical” values, weak on **heavy tail**

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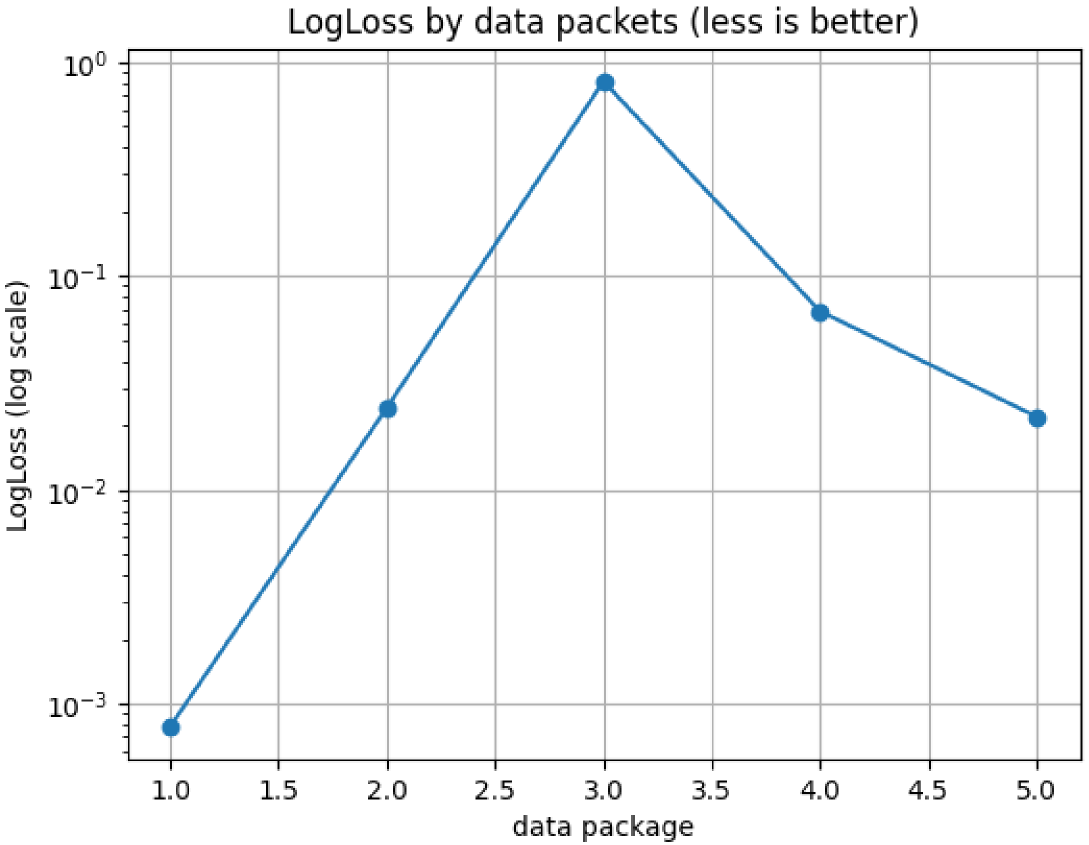
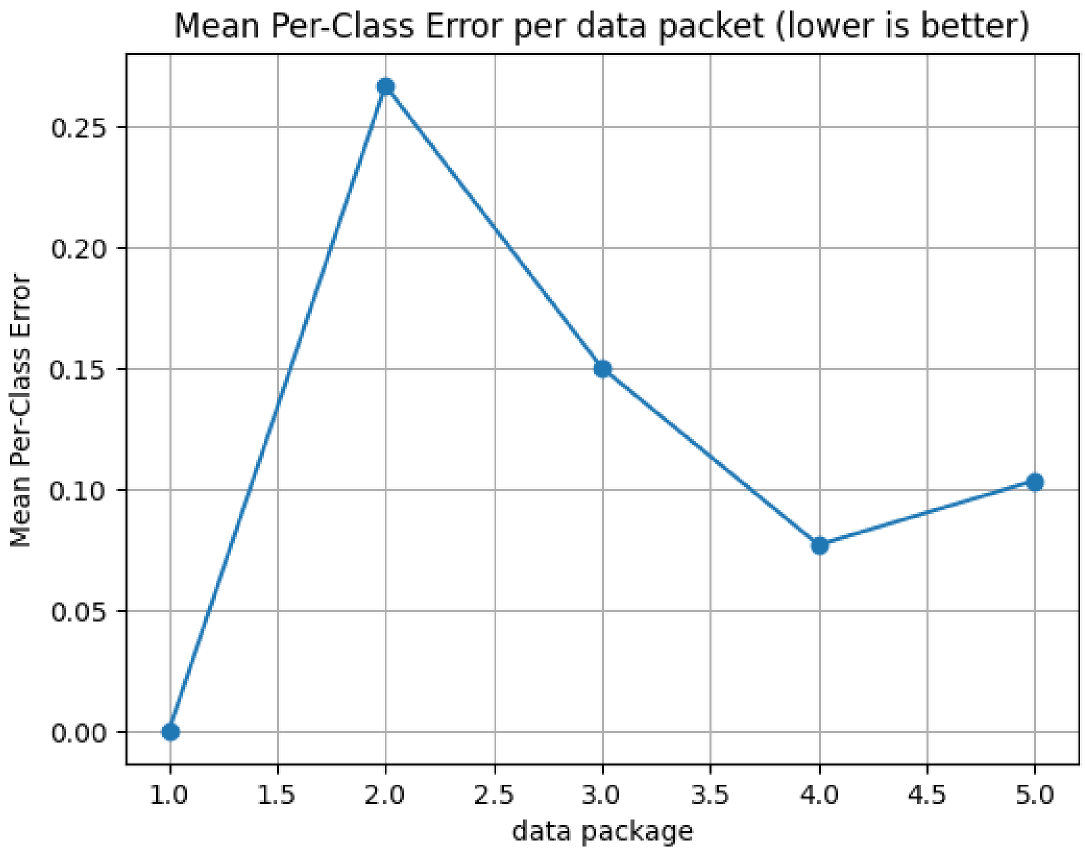
Thank you for your attention

Framework Workflow

1. Divide the data into training and testing sets **(B1–B5; Syn100/Syn500)**
2. Baseline, SMOTE, ADASYN, and ROSE techniques **(train-only; after TF-IDF)**.
3. GBM, XGBM, and RF, with **fixed settings**.
4. Calibration, ensure recall is greater than or equal to R_0 while maximising precision.
5. Choosing the optimal data and model combination, taking into account the **Recall >=0.98** and based on the PR-AUC metric **(+FP/1000 check).**
6. Stages B1 to B5 using both A) Incremental and B) Full Retrain **(binary: B2–B5)**.
7. Binary outcome by ensuring threshold stability, FN/FP profile **(+OOF aggregates)**.
8. For typing, implement multiclass strategies, considering both Macro and Weighted Recall **(baseline only)**.
9. Select the final mode **(+final model config)**.

Model Selection with H2O AutoML





Data Package	Leader algorithm	MSE	RMSE	LogLoss	Mean per-class error	HitRatio
1	**XGB**	0.000089	0.009451	0.000785	0.000000	1.000000
2	**GBM**	0.001659	0.040734	0.024234	0.266667	0.998006
3	**GBM**	0.300184	0.547890	0.811208	0.150000	0.998004
4	**GBM**	0.005220	0.072252	0.068447	0.076923	0.998504
5	**XGB**	0.001907	0.043664	0.022020	0.103448	0.998504

Binary model Syn100

Mode	Variant	Model	ROC-AUC	PR-AUC	FP/1000
Full retrain	ADASYN	GBM	0.9920	0.9196	66.8317
Full retrain	Baseline	GBM	0.9764	0.9192	251.2376
Full retrain	SMOTE	GBM	0.9914	0.9242	67.3267
Incremental	ADASYN	GBM	0.9803	0.6701	94.3069
Incremental	Baseline	GBM	0.9728	0.8487	257.4257
Incremental	SMOTE	GBM	0.9854	0.7815	91.9554
Full retrain	ADASYN	RF	0.9960	0.9023	24.2574
Full retrain	Baseline	RF	0.9975	0.9685	26.3614
Full retrain	SMOTE	RF	0.9966	0.9322	23.8861
Incremental	ADASYN	RF	0.9907	0.8058	52.9703
Incremental	Baseline	RF	0.9928	0.8899	87.8713
Incremental	SMOTE	RF	0.9916	0.8422	63.7376
Full retrain	ADASYN	XGB	0.9977	0.9673	24.1337
Full retrain	Baseline	XGB	0.9956	0.9030	27.5990
Full retrain	**SMOTE**	**XGB**	**0.9977**	**0.9696**	**22.0297**
Incremental	ADASYN	XGB	0.9949	0.8686	33.5396
Incremental	Baseline	XGB	0.9899	0.8387	50.7426
Incremental	SMOTE	XGB	0.9958	0.8924	31.8069

Binary model Syn500

Mode	Variant	Model	ROC-AUC	PR-AUC	Recall	Precision@R	FP/1000
Full retrain	**Baseline**	**GBM**	**0.704**	**0.312**	**0.792**	**0.062**	**588.2**
Incremental	Baseline	GBM	0.703	0.315	0.795	0.063	580.3
Full retrain	Baseline	RF	0.710	0.378	0.998	0.049	948.7
Incremental	Baseline	RF	0.708	0.359	0.990	0.049	924.3
Full retrain	Baseline	XGB	0.687	0.306	0.779	0.061	582.6
Incremental	Baseline	XGB	0.684	0.279	0.762	0.062	562.2
Full retrain	ROSE	GBM	0.643	0.223	1.000	0.049	951.5
Incremental	ROSE	GBM	0.626	0.151	1.000	0.049	951.5
Full retrain	ROSE	RF	0.530	0.071	1.000	0.049	951.5
Incremental	ROSE	RF	0.502	0.049	1.000	0.049	951.5
Full retrain	ROSE	XGB	0.640	0.205	1.000	0.049	951.5
Incremental	ROSE	XGB	0.613	0.136	1.000	0.049	951.5

Out of Fold and Platt calibration Syn100

Variant	Model	ROC-AUC	PR-AUC	Precision@R	Recall	FP/1000	Brier raw	Brier Platt	Platt-raw
ADASYN	GBM	0.99992	0.99373	0.96861	1.00000	0.49505	0.0123	0.0138	+0.0015
ADASYN	RF	0.99996	0.99752	0.97949	1.00000	0.39604	0.0077	0.0040	-0.0037
ADASYN	XGB	0.99995	0.99717	0.94544	1.00000	0.99010	0.0074	0.0063	-0.0011
Baseline	GBM	0.99998	0.99908	0.97754	1.00000	0.39604	0.0047	0.0042	-0.0005
Baseline	RF	0.99998	0.99878	0.97500	1.00000	0.49505	0.0053	0.0087	+0.0034
Baseline	XGB	0.99999	0.99931	0.97999	1.00000	0.29703	0.0045	0.0038	-0.0007
SMOTE	GBM	1.00000	0.99984	0.99444	1.00000	0.09901	0.0192	0.0198	+0.0006
SMOTE	RF	0.99997	0.99795	0.97949	1.00000	0.39604	0.0072	0.0038	-0.0034
SMOTE	XGB	0.99999	0.99912	0.98799	1.00000	0.19802	0.0099	0.0135	+0.0035

Out of Fold and Platt calibration Syn500

Variant	Model	ROC-AUC	PR-AUC	Precision@R	Recall	FP/1000	Brier raw	Brier Platt	Platt-raw
Baseline	GBM	0.83488	0.45761	0.06941	0.98050	638.65347	0.1193	0.0393	-0.0800
Baseline	RF	0.79676	0.44866	0.05537	0.98133	803.16832	0.0377	0.0366	-0.0011
Baseline	XGB	0.83489	0.46955	0.06881	0.98050	643.82178	0.1017	0.0397	-0.0619
ROSE	GBM	0.83084	0.47165	0.06782	0.98050	650.97030	0.9291	0.9492	+0.0201
ROSE	RF	0.79611	0.45045	0.05414	0.98299	820.81188	0.9474	0.9491	+0.0017
ROSE	XGB	0.83988	0.48527	0.06963	0.98050	631.62376	0.9501	0.9502	+0.0001